

CP2.3 — Methods in Calculus

1. Improper Integrals

An integral is **improper** if either:

- One or both limits are infinite: $\int_1^{\infty} f(x) dx$
- The integrand has a discontinuity (vertical asymptote) within the interval: $\int_0^1 \frac{1}{\sqrt{x}} dx$

We evaluate by replacing the problematic limit with a parameter and taking a limit.

1.1 Infinite Limits

$$\int_a^{\infty} f(x) dx = \lim_{t \rightarrow \infty} \int_a^t f(x) dx$$

If this limit exists and is finite, the integral **converges**. Otherwise it **diverges**.

Example: Determine whether $\int_1^{\infty} \frac{1}{x^2} dx$ converges.

$$\int_1^t \frac{1}{x^2} dx = \left[-\frac{1}{x} \right]_1^t = -\frac{1}{t} + 1 \xrightarrow{t \rightarrow \infty} 1$$

Converges to 1.

Example: Determine whether $\int_1^{\infty} \frac{1}{x} dx$ converges.

$$\int_1^t \frac{1}{x} dx = [\ln x]_1^t = \ln t \xrightarrow{t \rightarrow \infty} \infty$$

Diverges.

1.2 The p-Test (Critical Result)

$$\int_1^{\infty} \frac{1}{x^p} dx \text{ converges if and only if } p > 1$$

$$\int_0^1 \frac{1}{x^p} dx \text{ converges if and only if } p < 1$$

Proof for $\int_1^{\infty} x^{-p} dx$:

For $p \neq 1$: $\int_1^t x^{-p} dx = \left[\frac{x^{-p+1}}{1-p} \right]_1^t = \frac{t^{1-p} - 1}{1-p}$

- If $p > 1$: $1-p < 0$, so $t^{1-p} \rightarrow 0$ as $t \rightarrow \infty$. Converges to $\frac{1}{p-1}$.
- If $p < 1$: $1-p > 0$, so $t^{1-p} \rightarrow \infty$. Diverges.
- If $p = 1$: gives $\ln t \rightarrow \infty$. Diverges.

1.3 Discontinuities within Interval

$$\int_a^b f(x) dx = \lim_{\varepsilon \rightarrow 0^+} \int_a^{b-\varepsilon} f(x) dx \quad \text{if } f \text{ has singularity at } b$$

Example: Evaluate $\int_0^1 \frac{1}{\sqrt{x}} dx$.

$$\int_{\varepsilon}^1 x^{-1/2} dx = [2\sqrt{x}]_{\varepsilon}^1 = 2 - 2\sqrt{\varepsilon} \xrightarrow{\varepsilon \rightarrow 0^+} 2$$

Converges to 2.

Exam tip: Always write the integral with a limit parameter, evaluate the definite integral, then explicitly take the limit. Don't just "put in ∞ " — show the limiting process.

2. Mean Value of a Function

The **mean value** of $f(x)$ on the interval $[a, b]$ is:

$$\bar{f} = \frac{1}{b-a} \int_a^b f(x) dx$$

Geometrically, this is the height of a rectangle of width $(b-a)$ having the same area as the area under $f(x)$ from a to b .

Example: Find the mean value of $f(x) = x^2 + 1$ on $[1, 4]$.

$$\begin{aligned} \bar{f} &= \frac{1}{4-1} \int_1^4 (x^2 + 1) dx = \frac{1}{3} \left[\frac{x^3}{3} + x \right]_1^4 \\ &= \frac{1}{3} \left[\left(\frac{64}{3} + 4 \right) - \left(\frac{1}{3} + 1 \right) \right] \\ &= \frac{1}{3} \left(\frac{64}{3} + 4 - \frac{1}{3} - 1 \right) = \frac{1}{3} \cdot (21 + 3) = \frac{1}{3} \cdot 24 = 8 \end{aligned}$$

Example: Find the mean value of $f(x) = \sin x$ on $[0, \pi]$.

$$\bar{f} = \frac{1}{\pi} \int_0^{\pi} \sin x \, dx = \frac{1}{\pi} [-\cos x]_0^{\pi} = \frac{1}{\pi} (1 + 1) = \frac{2}{\pi}$$

3. Differentiation of Inverse Trigonometric Functions

$$\frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}, \quad |x| < 1$$

$$\frac{d}{dx}(\arccos x) = \frac{-1}{\sqrt{1-x^2}}, \quad |x| < 1$$

$$\frac{d}{dx}(\arctan x) = \frac{1}{1+x^2}$$

3.1 Proof for arcsin

Let $y = \arcsin x$, so $\sin y = x$. Differentiating implicitly:

$$\cos y \cdot \frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\sqrt{1-\sin^2 y}} = \frac{1}{\sqrt{1-x^2}}$$

(Taking positive root since $-\pi/2 \leq y \leq \pi/2$ so $\cos y \geq 0$.)

3.2 Chain Rule Extensions

Function	Derivative
$\arcsin\left(\frac{x}{a}\right)$	$\frac{1}{\sqrt{a^2 - x^2}}$
$\arctan\left(\frac{x}{a}\right)$	$\frac{a}{a^2 + x^2}$

$$\arcsin(f(x))$$

$$\frac{f'(x)}{\sqrt{1 - (f(x))^2}}$$

Example: Differentiate $y = \arctan(3x^2)$.

$$\frac{dy}{dx} = \frac{6x}{1 + (3x^2)^2} = \frac{6x}{1 + 9x^4}$$

4. Integration Using Inverse Trig Functions

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \arcsin\left(\frac{x}{a}\right) + C, \quad |x| < a$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C$$

Example: Evaluate $\int_0^1 \frac{1}{\sqrt{4 - x^2}} dx$.

$$a = 2. \left[\arcsin \frac{x}{2} \right]_0^1 = \arcsin \frac{1}{2} - \arcsin 0 = \frac{\pi}{6} - 0 = \frac{\pi}{6}$$

Example: Evaluate $\int \frac{1}{9 + x^2} dx$.

$$a = 3. \frac{1}{3} \arctan\left(\frac{x}{3}\right) + C$$

4.1 Completing the Square

When the denominator is a quadratic $ax^2 + bx + c$, complete the square first to write it in the form $(x - h)^2 + k^2$ or $k^2 - (x - h)^2$.

Example: Find $\int \frac{1}{x^2 + 4x + 13} dx$.

Complete the square: $x^2 + 4x + 13 = (x + 2)^2 + 9$.

Substitute $u = x + 2$:

$$\int \frac{du}{u^2 + 9} = \frac{1}{3} \arctan \frac{u}{3} + C = \frac{1}{3} \arctan \frac{x + 2}{3} + C$$

Example: Find $\int \frac{1}{\sqrt{8 - 2x - x^2}} dx$.

Complete the square: $8 - 2x - x^2 = 9 - (x + 1)^2$.

Substitute $u = x + 1$:

$$\int \frac{du}{\sqrt{9 - u^2}} = \arcsin \frac{u}{3} + C = \arcsin \frac{x + 1}{3} + C$$

Exam tip: Always check whether the numerator is a constant or has a variable part. If the numerator contains x , split as $\frac{Ax+B}{ax^2+bx+c} = \frac{A}{2a} \cdot \frac{2ax+b}{ax^2+bx+c} + \frac{\text{remainder}}{ax^2+bx+c}$. The first part integrates as a logarithm, the second as arctan.

5. Partial Fractions — Repeated and Irreducible Quadratic Factors

5.1 Repeated Linear Factors

$$\frac{f(x)}{(ax + b)^2(cx + d)} = \frac{A}{ax + b} + \frac{B}{(ax + b)^2} + \frac{C}{cx + d}$$

Example: Decompose $\frac{3x + 1}{(x - 1)^2(x + 2)}$.

$$\frac{3x+1}{(x-1)^2(x+2)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+2}$$

Multiply through: $3x+1 = A(x-1)(x+2) + B(x+2) + C(x-1)^2$

$$x=1: 4 = 3B \Rightarrow B = \frac{4}{3}$$

$$x=-2: -5 = 9C \Rightarrow C = -\frac{5}{9}$$

$$x=0: 1 = -2A + 2B + C \Rightarrow 1 = -2A + \frac{8}{3} - \frac{5}{9} \rightarrow 2A = \frac{8}{3} - \frac{5}{9} - 1 = \frac{24-5-9}{9} = \frac{10}{9} \rightarrow A = \frac{5}{9}$$

Check: $A + C = 0$ (coefficient of x^2 must be 0) ✓

5.2 Irreducible Quadratic Factors

$$\frac{f(x)}{(ax^2+bx+c)(dx+e)} = \frac{Ax+B}{ax^2+bx+c} + \frac{C}{dx+e}$$

where ax^2+bx+c does not factorise over $\mathbb{R}$.

Example: Decompose $\frac{5x^2+4}{(x^2+4)(x-1)}$.

$$\frac{5x^2+4}{(x^2+4)(x-1)} = \frac{Ax+B}{x^2+4} + \frac{C}{x-1}$$

Multiply through: $5x^2+4 = (Ax+B)(x-1) + C(x^2+4)$

$$x=1: 9 = 5C \Rightarrow C = \frac{9}{5}$$

Expand and compare coefficients:

$$x^2: 5 = A + C \Rightarrow A = 5 - \frac{9}{5} = \frac{16}{5}$$

$$x^0: 4 = -B + 4C = -B + \frac{36}{5} \Rightarrow B = \frac{36}{5} - 4 = \frac{16}{5}$$

Check $x^1: 0 = B - A = 0$ ✓

6. Integration by Parts — Repeated Application (Tabular Method)

$$\int u dv = uv - \int v du$$

For integrals requiring repeated integration by parts (e.g., $\int x^n e^x dx$, $\int e^{ax} \sin(bx) dx$), the **tabular method** (also called the DI method) is efficient.

6.1 Tabular Method for Polynomial × Exponential

Example: Find $\int x^3 e^x dx$ using the tabular method.

Derivatives (D)	Integrals (I)	Sign
x^3	e^x	+
$3x^2$	e^x	−
$6x$	e^x	+
6	e^x	−
0	e^x	+

Multiply diagonally with alternating signs:

$$\int x^3 e^x dx = x^3 e^x - 3x^2 e^x + 6x e^x - 6e^x + C = e^x(x^3 - 3x^2 + 6x - 6) + C$$

6.2 Cyclic Integration by Parts

Example: Find $\int e^{2x} \cos 3x dx$.

Let $I = \int e^{2x} \cos 3x dx$.

First IBP: $u = e^{2x}$, $dv = \cos 3x \, dx$, so $v = \frac{\sin 3x}{3}$:

$$I = \frac{e^{2x} \sin 3x}{3} - \int \frac{2e^{2x} \sin 3x}{3} \, dx$$

Second IBP on remaining integral: $u = e^{2x}$, $dv = \sin 3x \, dx$, $v = -\frac{\cos 3x}{3}$:

$$\int e^{2x} \sin 3x \, dx = -\frac{e^{2x} \cos 3x}{3} + \int \frac{2e^{2x} \cos 3x}{3} \, dx = -\frac{e^{2x} \cos 3x}{3} + \frac{2I}{3}$$

Substitute back:

$$I = \frac{e^{2x} \sin 3x}{3} - \frac{2}{3} \left(-\frac{e^{2x} \cos 3x}{3} + \frac{2I}{3} \right)$$

$$I = \frac{e^{2x} \sin 3x}{3} + \frac{2e^{2x} \cos 3x}{9} - \frac{4I}{9}$$

$$\frac{13I}{9} = \frac{3e^{2x} \sin 3x + 2e^{2x} \cos 3x}{9}$$

$$I = \frac{e^{2x}(3 \sin 3x + 2 \cos 3x)}{13} + C$$

Exam tip: For cyclic IBP ($e^{ax} \sin bx$ type), always call the original integral I , apply IBP twice keeping the same choice of "u", then solve for I . Don't forget the constant of integration.

7. Summary of Integration Formulas

Integrand	Result
$\frac{1}{\sqrt{a^2 - x^2}}$	$\arcsin\left(\frac{x}{a}\right) + C$
$\frac{1}{a^2 + x^2}$	$\frac{1}{a} \arctan\left(\frac{x}{a}\right) + C$
$\frac{1}{\sqrt{x^2 + a^2}}$	$\operatorname{arsinh}\left(\frac{x}{a}\right) + C = \ln(x + \sqrt{x^2 + a^2}) + C$

$\frac{1}{\sqrt{x^2 - a^2}}$	$\operatorname{arcosh}\left(\frac{x}{a}\right) + C = \ln(x + \sqrt{x^2 - a^2}) + C$
$\frac{1}{a^2 - x^2}$	$\frac{1}{2a} \ln\left \frac{a+x}{a-x}\right + C$

8. Exam-Style Practice Questions

- [Improper integral]** Determine whether $\int_0^{\infty} x e^{-x} dx$ converges, and if so, find its value. (Hint: use integration by parts.)
- [Improper integral]** Show that $\int_0^1 \frac{1}{x^{2/3}} dx$ converges and find its value.
- [Mean value]** Find the mean value of $f(x) = e^{2x}$ over the interval $[0, 3]$.
- [Inverse trig integration]** Find $\int \frac{3}{\sqrt{16 - 9x^2}} dx$.
- [Completing the square]** Evaluate $\int_0^1 \frac{2x + 3}{x^2 + 2x + 5} dx$.
- [Partial fractions]** Show that $\int \frac{2x^2 + 2x + 1}{(x^2 + 1)(2x - 1)} dx = \ln|2x - 1| + \arctan x + C$.
- [IBP tabular]** Find $\int x^2 \sin x dx$.
- [Cyclic IBP]** Find $\int e^{-x} \sin 2x dx$.

General exam tips for CP2.3:

- Improper integrals: always write $\lim_{t \rightarrow \infty}$ explicitly and evaluate before taking the limit
- For inverse trig integrals, identify a correctly: in $\frac{1}{a^2 + x^2}$, a is the square root of the constant
- Completing the square: write out every algebraic step — errors here lose all subsequent marks
- IBP cyclic: rearranging to get I on one side is the key step — don't omit the $+C$ at the end